

Report R22 — The External Probe on Five Encoders, and a Single Split Quoted as a Number

The external site signature clears its null in all five encoders. R20’s headline contrast was one cross-validation split, and it was an unlucky one.

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Summary

R20 measured EAGLE against HTMC^P on two encoders and reported Phikon-v2 1.0000 against dinov2-large 0.8768, concluding the signature was encoder-independent in existence but not in magnitude. R21 then showed that two-point claims in this series have twice been wrong. This was the last two-point claim left, so it is now run on all five encoders.

Verdict: EXTERNAL_SIGNATURE_PRESENT_MAGNITUDE_VARIES, from R20’s own thresholds, hashed before R20’s arms were read. Every encoder separates two institutions far above its permutation null at $p = 0.0050$.

Encoder	Corpus	dim	SET_HE	25 splits: mean ± sd	range	null
owkin/phikon-v2	histology	1024	1.0000	1.0000 ± 0.0000	1.0000–1.0000	0.5023
facebook/dinov2-large	natural	1024	0.8768	0.9176 ± 0.0247	0.8583–0.9528	0.4990
MahmoodLab/UNI	histology	1024	1.0000	0.9937 ± 0.0110	0.9630–1.0000	0.5003
bioptimus/H-optimus-0	histology	1536	1.0000	0.9961 ± 0.0078	0.9713–1.0000	0.4923
paige-ai/Virchow2	histology	1280	0.9528	0.9601 ± 0.0175	0.9157–0.9815	0.4965

1 The gate that halted, and what it found

A first attempt at this run used a fresh seed. The reproduction gate compared its dinov2-large value against R20’s 0.8768, got **0.9240**, and halted.

That is not a coding difference. The seed sets the **StratifiedKFold** split, and on 76 slides with roughly fifteen per fold a couple of reassignments move balanced accuracy by several points. **R20’s 0.8768 was one draw from a distribution with a standard deviation of 0.0247**, and it sat near the bottom of it: the 25-split mean is 0.9176 and the observed minimum 0.8583.

So R20’s substantive conclusion survives — the signature is present under every encoder and total only under some — but its headline contrast was overstated. The honest gap between Phikon-v2 and dinov2-large on these slides is **1.000 versus 0.918**, not 1.000 versus 0.877. **R20 is corrected.**

What this run should have quoted from the start is the distribution, not a point. Single-split balanced accuracy on 76 samples is a noisy statistic and there was never a reason to treat it as stable. The single-seed column is retained only to demonstrate exact reproduction.

2 What the five encoders show

The signature is universal. All five clear their nulls decisively, every null sits within 0.008 of chance, and every p is at the floor of 0.0050 for 200 permutations.

Phikon-v2 is genuinely saturated — 1.0000 on all 25 splits, standard deviation exactly zero. That is not a lucky split; two institutions are perfectly linearly separable in its mean-pooled feature space.

The histology encoders cluster high and the natural-image encoder sits below them: means 0.9601, 0.9937, 0.9961 and 1.0000 against 0.9176. The ordering matches R21’s finding on methylation, where the corpus split was much starker. Here all five are high, so the corpus effect is a difference of degree rather than of kind, and it should not be leaned on — with four histology encoders spanning 0.960 to 1.000 and one natural-image encoder at 0.918, a single non-pathology point is not enough to attribute the gap to corpus.

Feature width is a confound on magnitude. H-optimus-0 has 1536 dimensions and Virchow2 1280 against 1024 for the rest. A wider representation gives a linear probe more room on 76 samples. H-optimus-0 is near ceiling and Virchow2 is not, so width plainly does not order the results — but it is not controlled either.

3 SET_ALL, and the stain shortcut in the open

Mixing the archive’s 53 immunohistochemistry slides with the 76 H&E ones raises dinov2-large from 0.8768 to 0.9444, exactly as R20 found. Adding a different stain makes the task easier, which is the shortcut R12’s audit predicted in writing. SET_ALL is reported as an upper bound and is not a measurement of site signal.

4 Limitations

Two institutions, 76 H&E slides. This bounds existence, not effect size, and no number here is a powered external benchmark.

Slide-level mean features, not the attention-MIL representation.

Stain from filename suffixes (HE / CHR / P16), the only stain field available; source mislabelling propagates.

One natural-image encoder. The corpus ordering in §3 rests on a single non-pathology point and is stated as an observation, not a result.

Ceiling effects make three of the five incomparable. Phikon-v2, UNI and H-optimus-0 all touch 1.0000. Differences among saturated measurements are not interpretable, which is a reason to regard the whole external probe as a presence test rather than a ranking.

Two of five are CC-BY-NC-ND-4.0 (UNI, Virchow2). Licences are recorded in the feature files.

Reproducing this report

evidence/m19_external_probe_five.py (the whole run), evidence/config.yaml (pre-declared rules, SHA-256 025add55f381b264...), evidence/gates.json (5 gates, including the one that halted the first attempt), evidence/results.json with set_he_seed_distribution per encoder, evidence/log.jsonl, and the five $129 \times d$ mean-embedding matrices with cohort and H&E flags inline.